Statistical inference using normal distributions

Overview

Review and continuation of probability distributions and normal distributions

Confidence intervals for a proportion using normal distributions

Hypothesis tests for a proportion using normal distributions

If there is time: more practice!

Review and continuation of using normal distributions for inference

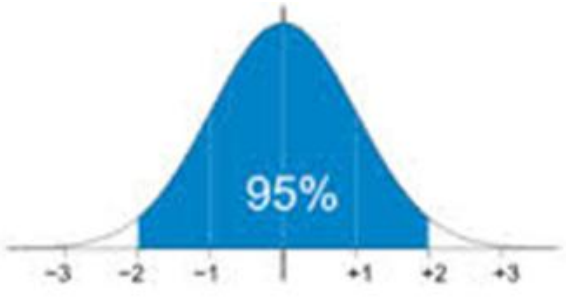
Review: Normal probability distributions

Normal distributions are a family of bell-shaped curves with two parameters

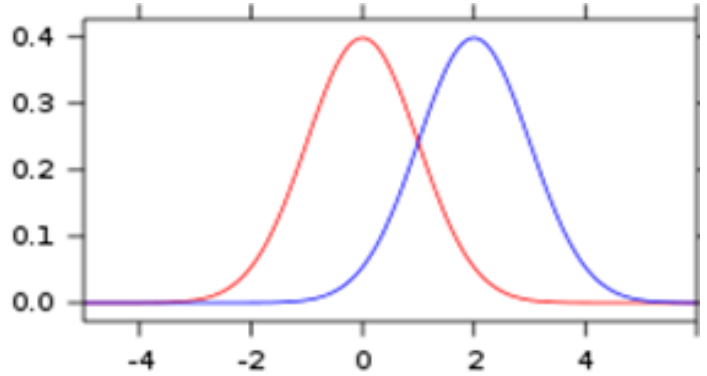
- The mean: μ
- The standard deviation: σ

Once μ and σ are set, we have a specific probability distribution that we use to get probabilities of events, generate random data, etc.

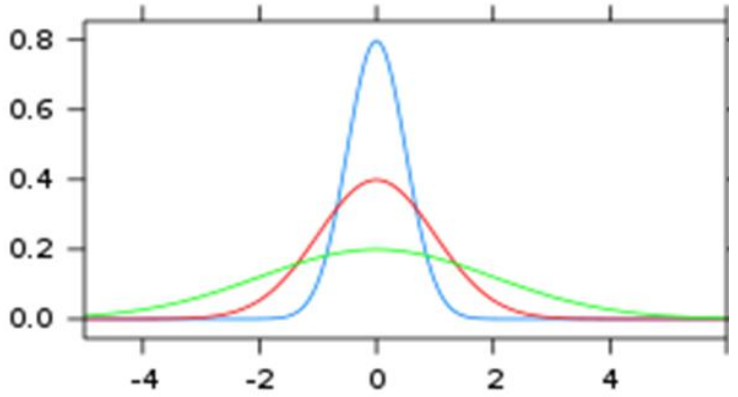




Changing μ



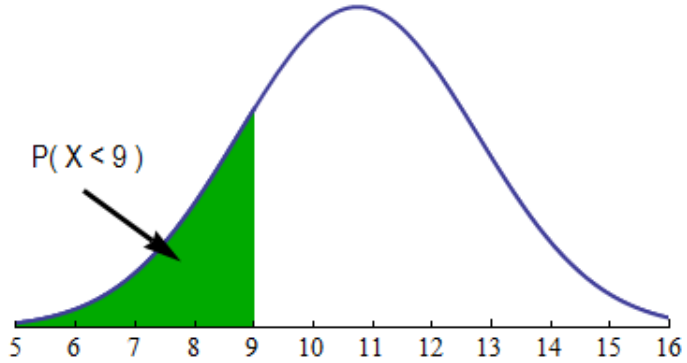
Changing σ



Review: Normal probability functions

Generate random data: X ~ N(μ, σ)

rnorm(num_points, mean, sd)



Plot the density curve

plot_norm(mean, sd)

- The total area under the curve f(x) is equal to 1
- The curve is always ≥ 0

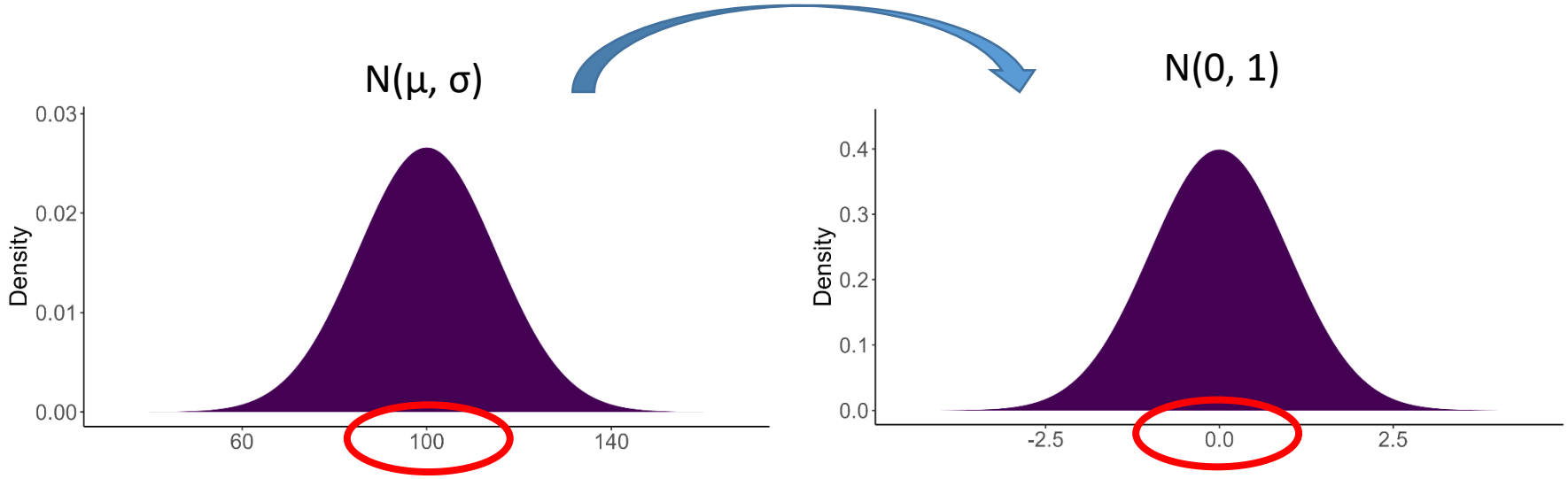
Get the probability that we would get a random value less than x: P(X < x)

pnorm(x_val, mean, sd)

Standard Normal N(0, 1)

Since all normal distributions have the same shape, it is convenient to convert them to a standard scale with: **$\mu = 0$, $\sigma = 1$**

This is called the **standard normal** distribution: $Z \sim N(0, 1)$



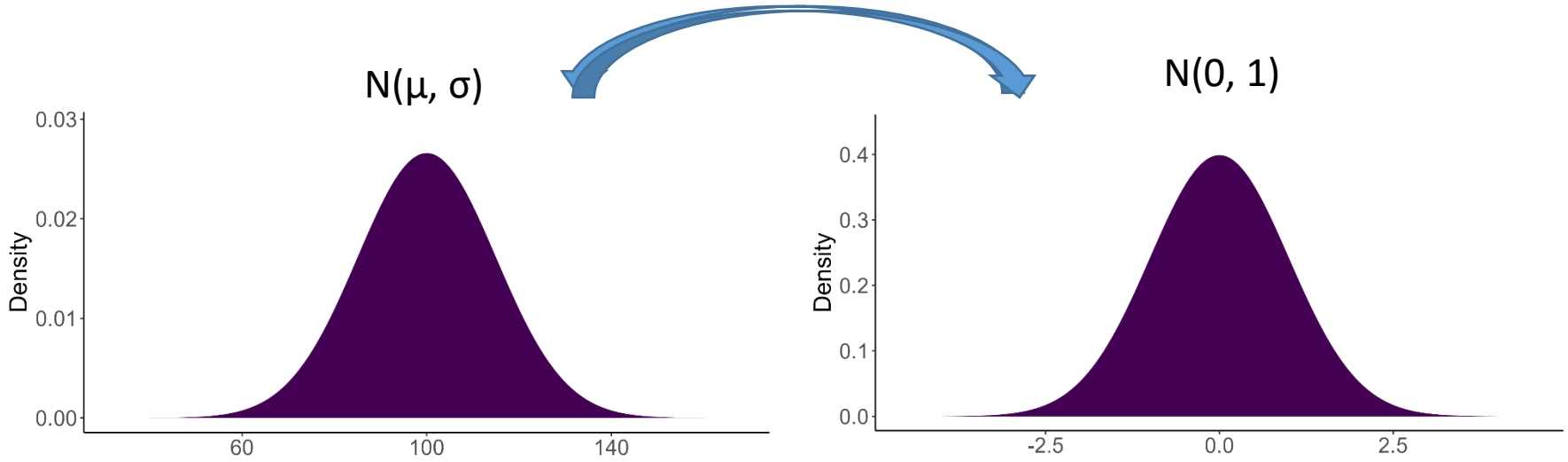
Converting to the standard normal distribution

We can apply a z-score transformation to any normally distributed random variable $X \sim N(\mu, \sigma)$ to convert it to the standard normal distribution $Z \sim N(0, 1)$:

$$Z = (X - \mu)/ \sigma$$

To convert from $Z \sim N(0, 1)$ to any $X \sim N(\mu, \sigma)$, we reverse the standardization with:

$$X = \mu + Z \cdot \sigma$$

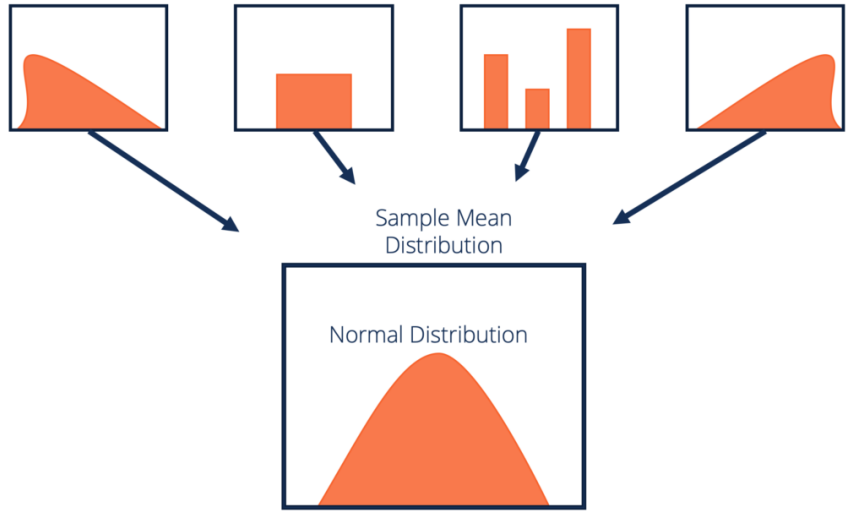


The Central Limit Theorem

Central Limit Theorem

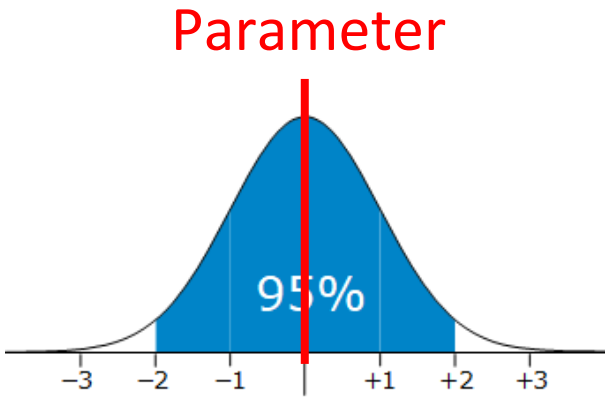
For random samples with a sufficiently large sample size (n), the distribution of sample statistics for a mean ($\bar{x}$) or a proportion ($\hat{p}$) is:

- Normally distributed
- Centered at the value of the population parameter



Upshot: We can create confidence intervals and run hypothesis tests using the normal distribution

- Rather than using computational methods like the bootstrap or randomization methods to create a null distribution!



Central Limit Theorem for Sample Proportions

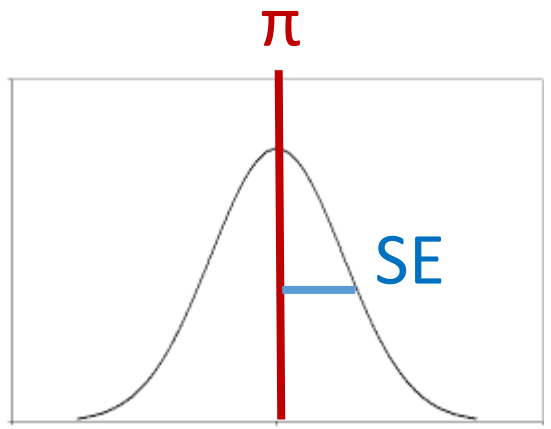
When taking samples of size n from a population with a proportion parameter π, the **sampling distribution** of p̂'s has the following characteristics:

Shape: If the sample size is sufficiently large, the distribution is reasonably normal

Center: The mean is equal to the population proportion π

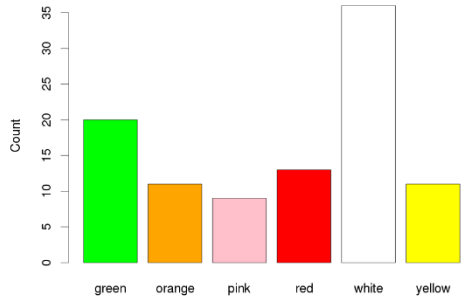
Spread: The standard error is: $SE = \sqrt{\frac{\pi(1-\pi)}{n}}$

$$\hat{P} \sim N(\pi, \sqrt{\frac{\pi(1-\pi)}{n}})$$



A normal distribution is a good approximation as long as:

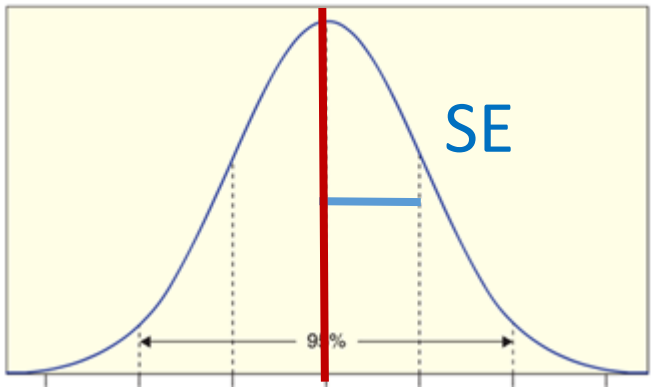
$n\pi \geq 10$ and $n(1-\pi) \geq 10$



π_{red}

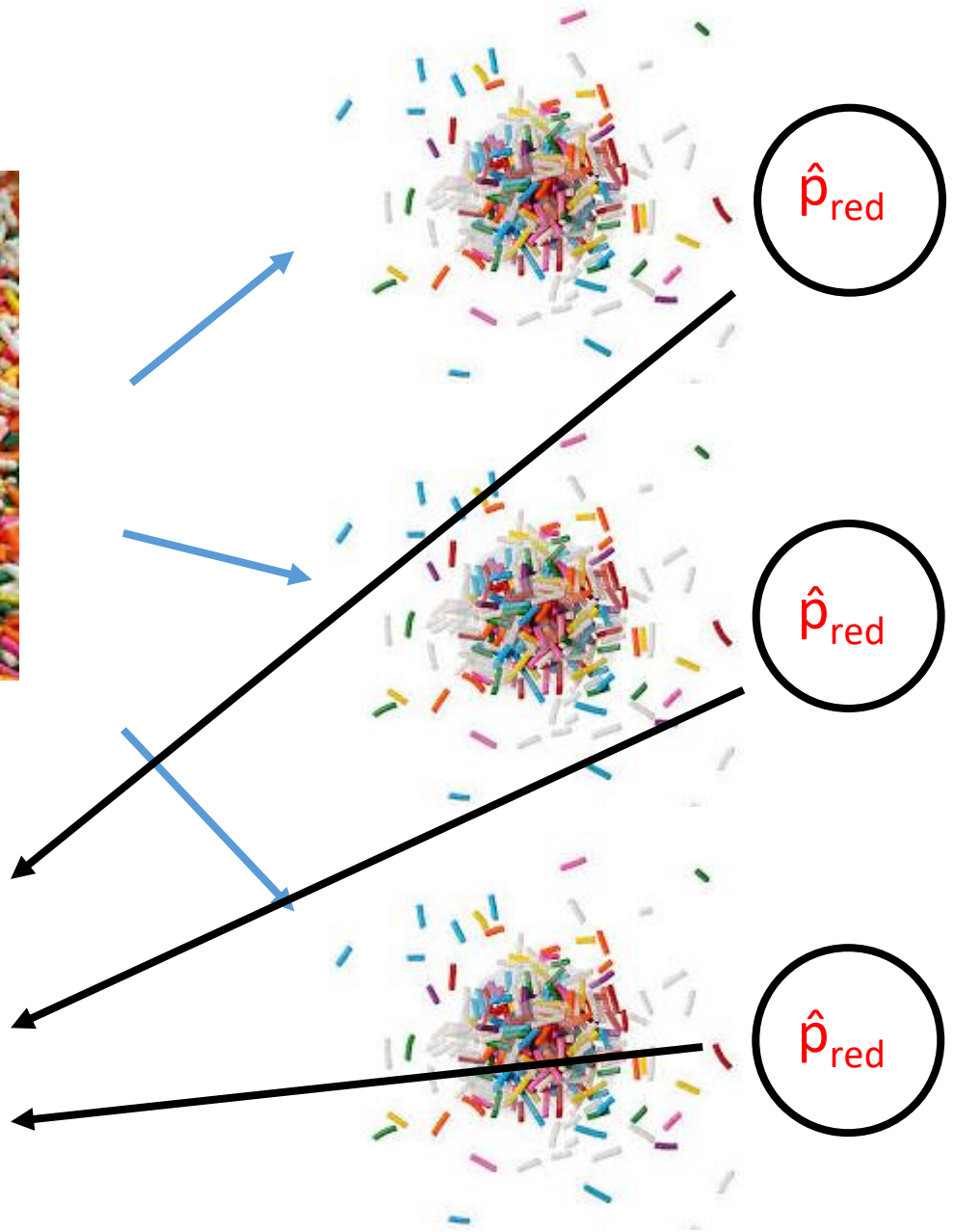


π



$$SE = \sqrt{\frac{\pi(1-\pi)}{n}}$$

$$\hat{P} \sim N(\pi, \sqrt{\frac{\pi(1-\pi)}{n}})$$



Sampling distribution!

The plan

For large n, the sampling distributions of $\hat{p}$ (and $\bar{x}$) have normal distributions

For confidence intervals, we can use formulas for the standard error

For hypothesis tests, we can convert any normal distribution $N(\mu, \sigma)$ into a standard normal distribution $N(0, 1)$ and then use this as our null distribution





Parametric confidence intervals for proportions

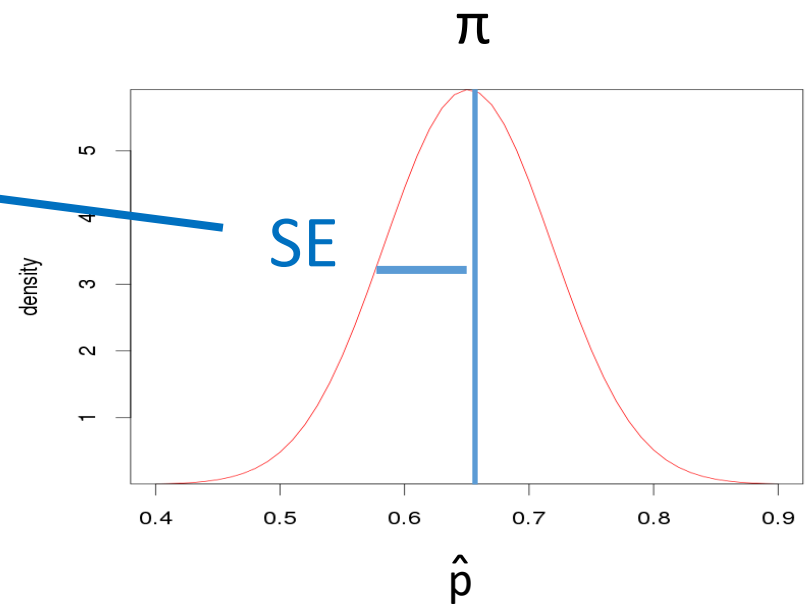
Summary: Central Limit Theorem for Sample Proportions

The Central Limit Theorem shows that the sampling distribution of proportions p̂ is normally distributed with the SE:

- (A normal distribution is a good approximation as long as: $n\pi \geq 10$ and $n(1-\pi) \geq 10$)

$$SE = \sqrt{\frac{\pi(1-\pi)}{n}}$$

$$\hat{P} \sim N(\pi, \sqrt{\frac{\pi(1-\pi)}{n}})$$



Can we use the formula for standard error to create confidence intervals?

- e.g., we can create a 95% confidence interval using: $\hat{p} \pm 2 \cdot SE$?

Standard Error for Sample Proportions

Note: we don't usually know π, so we can't compute the standard error exactly using the formula:

$$SE = \sqrt{\frac{\pi(1-\pi)}{n}}$$



However, we can substitute p̂ for π and then we can get an estimate of the standard error:

$$\hat{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$



Comparing formula SE to the bootstrap SE

In previous classes we have used the bootstrap to get an estimate of the standard error SE*

For example, if we had a sample of 100 sprinkles

We've gotten the SE* for the proportion of red sprinkles using…

```
bootstrap_dist  <-  do_it(100000) *  {
    boot_sample <- sample(my_sprinkles, replace = TRUE)
    get_proportion(boot_sample, 'red')
}

bootstrap_SE <- sd(bootstrap_dist)
```



Color
White
Red
Red
White
Green
White
.
.
.
White
Green

n = 100 sprinkles

Comparing formula SE to the bootstrap SE

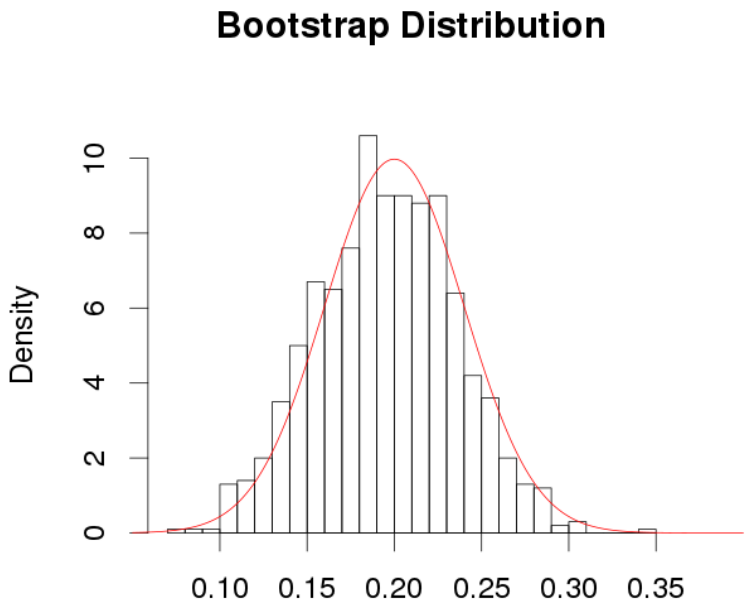
For my sprinkles I got:

- Bootstrap SE = 0.039959
- Formula SE = 0.04

$\hat{p} = 0.20$

$n = 100$

$$\hat{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$




```
SE <- sqrt(  (.2 * (1 - .2) ) /100  )
```

$$CI = \hat{p} \pm z^* \cdot SE$$

Confidence intervals for a single proportion

Suppose we have a sample of size n of categorical data

Suppose that n is large enough so that $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$

A confidence interval for a population proportion π can be computed from our random sample of size n using:

Equation for SE

$$\hat{p} \pm z^* \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Where $\hat{p}$ is the sample proportion and z* is a standard normal endpoint to give the desired confidence level

- i.e., we can get the z* value using the cnorm() function

One true love?

A survey asked 2625 people whether they agreed with the statement "There is only one true love for each person"

1812 of the respondents disagreed

Compute a 90% confidence interval for the proportion who disagreed

$$\hat{p} \pm z^* \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Let's try it in R!

One true love?

$$\hat{p} \pm z^* \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

n <- 2625

p_hat <- 1812/2625 = .69

SE <- sqrt((p_hat * (1 - p_hat))/n)

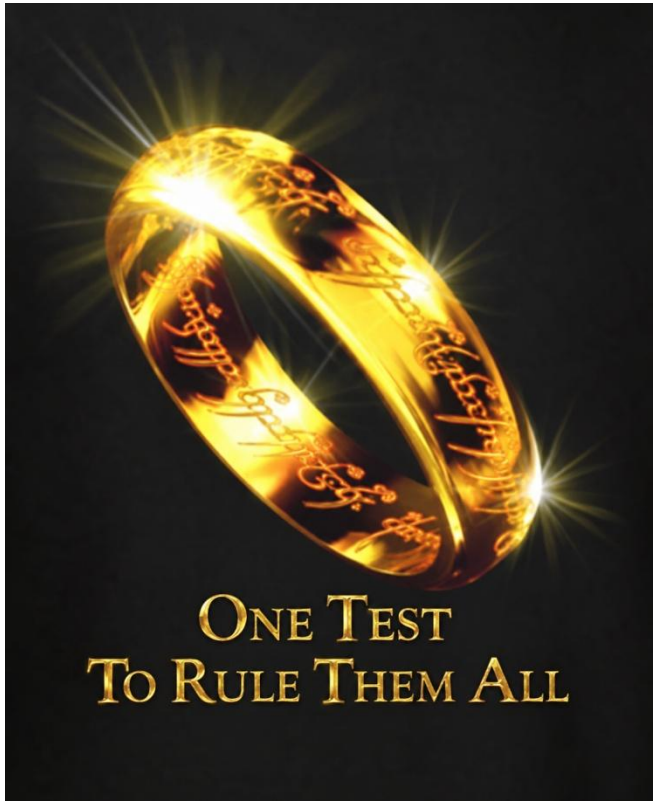
z_star <- cnorm(.90, 0, 1) = 1.64

ME <- z_star * SE = .0148

CI <- c(p_hat - ME, p_hat + ME) = [.675 .705]

Hypothesis tests for proportions using a normal distribution

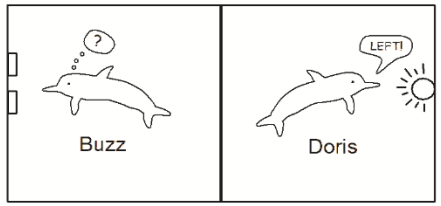
How many hypothesis tests are there?



Just follow the 5 hypothesis testing steps…

Randomization/computational test for proportions

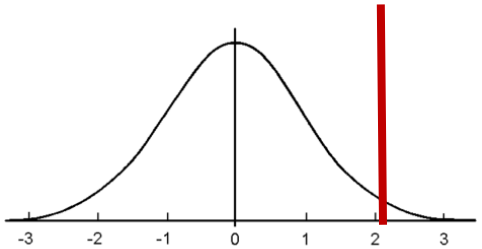
1. State the null and the alternative hypotheses
   - H_0: $\pi = 0.5$ (or another value such as 0.6, etc.)
   - H_A: $\pi > 0.5$





2. Calculate the observed statistic
   - $\hat{p} = .937$

3. Create a null distribution that is consistent with the null hypothesis
   - Null distribution is created by flipping coins: rflip(16, .5)/16



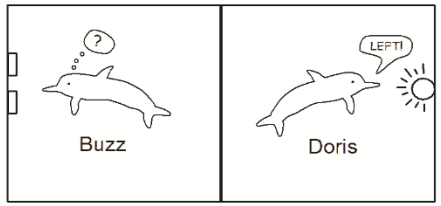
4. Examine how likely the observed statistic is to come from the null distribution
   - pval <- ptail(obs_stat, null_dist, lower.tail = FALSE)

5. Make a judgement
   - If the p-value is small (often less than 0.05) then we can reject the null hypothesis

Parametric hypothesis test for proportions

1. State the null and the alternative hypotheses
   - H_0: π = 0.5 (or another value such as 0.6, etc.)
   - H_A: π > 0.5

1. The same

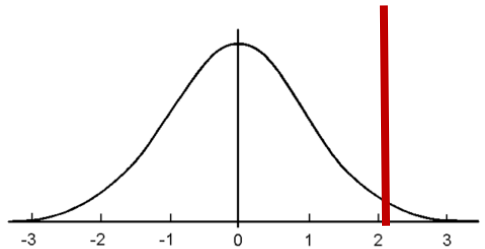




2. Calculate the observed statistic

$$z = \frac{\hat{p} - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$$

   - ~~p̂ = .937~~



3. Create a null distribution that is consistent with the null hypothesis
   - ~~Null distribution is created by flipping coins: rflip(16, .5)/16~~
   - Null distribution is the standard normal distribution: plot_norm(0, 1)

4. Examine how likely the observed statistic is to come from the null distribution
   - ~~pval <- ptail(obs_stat, null_dist, lower.tail = FALSE)~~ pval <- pnorm(z, 0, 1, lower.tail = FALSE)

5. Make a judgement
   - If the p-value is small (often less than 0.05) then we can reject the null hypothesis

5. The same

Five-step hypothesis test for proportions

What is our first step?

Step 1: State the null and alternative hypothesis

For proportions, what are our null and alternative hypotheses?

$H_0: \pi = \pi_0$

$H_A: \pi > \pi_0$

The same as when we use randomization tests

What do we do next…?

Step 2: Calculate the statistic of interest



When the null distribution is normal, it is often convenient to use a standard normal test statistic

$$z = \frac{Sample\ \ Statistic\ \ -\ \ Null\ \ Parameter}{SE} \qquad z = \frac{\hat{p} - \pi_0}{SE}$$

For proportions, our null hypothesis is: $H_0: \pi = \pi_0$

From the central limit theorem for proportions we know: $SE = \sqrt{\frac{\pi(1-\pi)}{n}}$

- What is the problem using this formula?

If the null hypothesis is true, $\pi = \pi_0$ so the correct π to use in this formula is π_0

Thus, our test statistic is: $z = \frac{\hat{p} - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$

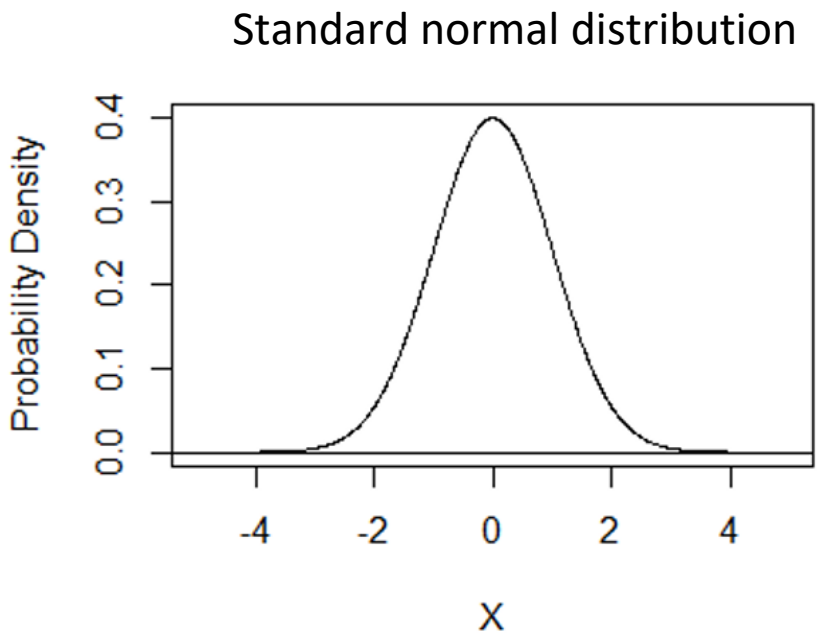
Step 3: Create/plot the null distribution

Our test statistic is: $z = \frac{\hat{p} - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$

Q: What is the appropriate null distribution?

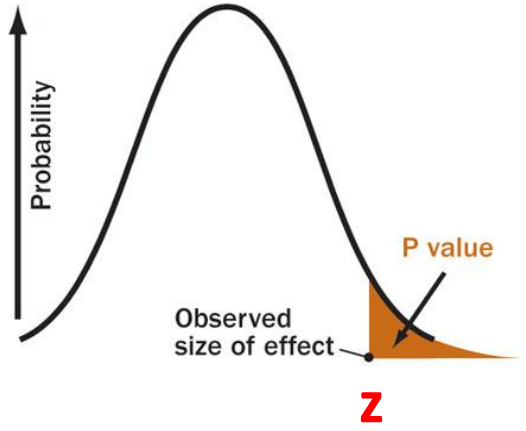
A: The standard normal distribution!

plot_norm(0, 1)



Step 4: Calculate the p-value

The p-value for the test is the probability a standard normal value is beyond this standardized test statistic



$$P(Z \geq z_{obs};\ \mu = 0,\ \sigma = 1)$$

```
pval <- pnorm(z, 0, 1, lower.tail = FALSE)
```

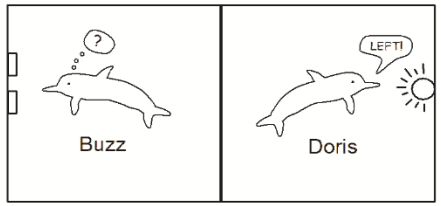
Step 5: Make a decision



The same as when we use randomization tests

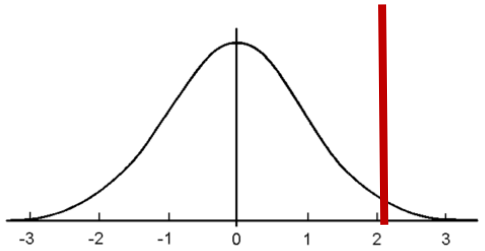
Summary parametric hypothesis test for proportions

1. State the null and the alternative hypotheses
   - H_0: π = 0.5 (or another value such as 0.6, etc.)
   - H_A: π > 0.5





2. Calculate a z-statistic $$z = \frac{\hat{p} - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$$



3. The null distribution is the standard normal distribution
   - We can visualize our null distribution using: plot_norm(0, 1)

4. Examine how likely the observed statistic is to come from the null distribution
   - pval <- pnorm(z_stat, 0, 1, lower.tail = FALSE)

5. Make a judgement
   - If the p-value is small (often less than 0.05) then we can reject the null hypothesis in favor of the alternative hypothesis

Do goalies guess the direction of a penalty shot less than 50% of the time?



1. Start by stating H_0 and H_A

H_0: $\pi = 0.5$

H_A: $\pi < 0.5$

2. Calculate the observed statistic

- Goalkeepers correctly guessed the direction 41% of the time out of 128 kicks

Can you compute a z-statistic?

Are the conditions for using a normal distribution met?

- $n\pi_0 \geq 10$ and $n(1-\pi_0) \geq 10$
- $128 \cdot 0.5 = 64$
- $128 \cdot (1 - 0.5) = 64$

z = -2.036

$$z = \frac{\hat{p} - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$$

0.41

0.5

128

Do goalies guess the direction of a penalty shot less than 50% of the time?

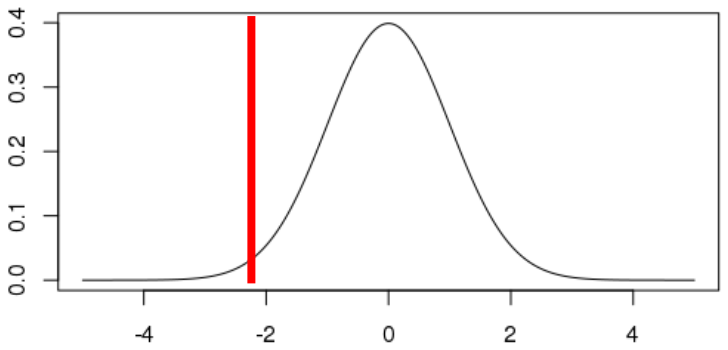
Steps: 3-4. What is the probability one would get a z-statistic as small or smaller than -2.036 from a standard normal distribution?

pnorm(-2.036, 0, 1)

Normal area app $P(X \leq x)$

p-value = 0.021

Standard normal null distribution



Step 5?



More practice

Sinister lawyers

10% of the American population is left-handed

A study found that out of a random sample of 105 lawyers, 16 were left-handed

Test whether the proportion of left-handed lawyers is greater than the proportion found in the American population.

1. State the null and alternative hypothesis

2-4. Calculate the statistic of interest and calculate the p-value

- Hint: the pnorm() function will be useful

5. What do you conclude?

$$z = \frac{\hat{p} - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$$

Sinister lawyers

1. State the null and alternative hypothesis
   - H_0: $\pi = .10$
   - H_A: $\pi > .10$

2-4. Calculate the statistic of interest and the p-value

$\hat{p} = 16/105 = .152$

SE = sqrt((.10 * (1 - .10))/105) = .029

z = (.152 - .10)/.029 = 1.79

pnorm(z, 0, 1, lower.tail = FALSE) = .037

$$z = \frac{\hat{p} - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}}$$



5. What do you conclude?